ECE341 HOMEWORK 5

Problem 1

State Table

Q_1	Q_0	S_b	D_1	D_0
0	0	0	0	0
0	0	1	0	1
0	1	0	1	0
0	1	1	0	1
1	0	0	0	0
1	0	1	1	1
1	1	0	1	0
1	1	1	0	1

K-maps and Derived Expressions

S_b \ Q_1Q_0	00	01	11	10
0	0	1	1	0
1	0	0	0	1

$D_1 = Q_0\overline{S_b} + Q_1\overline{Q_0}S_b$

S_b \ Q_1Q_0	00	01	11	10
0	0	0	0	0
1	1	1	1	1

$D_0 = S_b$

Since this is a Moore FSM, the output depends only on the current state. Z is only high when the machine is in state $S3$ ($Q_1Q_0 = 11$), or $Z = Q_1 \cdot Q_0$.

Problem 2

State Table

S_b \ Q_1Q_0	00	01	11	10
0	0	0	X	0
1	0	0	X	1

$D_1 = Q_1 S_b$

Q_1Q_0	$S_b=0$ $Q_1'Q_0'$	$S_b=1$ $Q_1'Q_0'$	$S_b=0$ Z	$S_b=1$ Z
00	00	01	0	0
01	10	01	0	0
10	00	01	0	1

K-maps and Derived Expressions

S_b \ Q_1Q_0	00	01	11	10
0	0	1	X	0
1	0	0	X	0

$D_1 = Q_1' = Q_0 \overline{S_b}$

S_b \ Q_1Q_0	00	01	11	10
0	0	0	X	0
1	1	1	X	1

$D_0 = Q_0' = S_b$

Problem 3

State Table

S_1S_0 \ Q_1Q_0	00	01	11	10
00	X	0	0	1
01	0	0	0	0
11	X	0	0	0
10	X	1	0	0

S_1S_0	Key
00	1
01	3
10	4
11	X

$$D_1 = Q_1' = Q_1\overline{Q_0}\,\overline{S_1}\,\overline{S_0} + \overline{Q_1}Q_0 S_1 \overline{S_0}$$

$$S_1 = \overline{Is1} \cdot \overline{Is3}$$

$$S_0 = \overline{Is1} \cdot \overline{Is4}$$

- Detection for 1 (0001): $Is_1 = \overline{K_3} \cdot \overline{K_2} \cdot \overline{K_1} \cdot K_0$
- Detection for 3 (0011): $Is_3 = \overline{K_3} \cdot \overline{K_2} \cdot K_1 \cdot K_0$
- Detection for 4 (0100): $Is_4 = \overline{K_3} \cdot K_2 \cdot \overline{K_1} \cdot \overline{K_0}$

K-maps and Derived Expressions

S_1S_0 \ Q_1Q_0	00	01	11	10
00	X	0	0	1
01	0	0	0	0
11	X	0	0	0
10	X	1	0	0

S_1S_0	Key
00	1
01	3
10	4
11	X

$$D_1 = Q_1' = Q_1\overline{Q_0}\,\overline{S_1}\,\overline{S_0} + \overline{Q_1}Q_0 S_1 \overline{S_0}$$

$$S_1 = \overline{Is1} \cdot \overline{Is3}$$

$$S_0 = \overline{Is1} \cdot \overline{Is4}$$

S_1S_0 \ Q_1Q_0	00	01	11	10
00	0	0	0	1
01	1	1	0	1
11	0	0	0	0
10	0	1	0	0

$$D_0 = Q_0' = \overline{Q_1}\,\overline{S_1}\,S_0 + Q_1\overline{Q_0}\,\overline{S_1}$$

Problem 4

State Table

Q_1Q_0 Current state	Key	$Q_1'Q_0'$ Next state	$Z(Q_1, Q_0, Key)$
00	2	01	0
00	≠2	00	0
01	0	10	0
01	2	01	0
01	≠0,2	00	0
10	2	11	0
10	≠2	00	0
11	5	00	1
11	0	10	0
11	2	01	0
11	≠5,0,2	00	0

- Detection for 0 (0000): $Is_1 = \overline{K_3} \cdot \overline{K_2} \cdot \overline{K_1} \cdot \overline{K_0}$
- Detection for 2 (0010): $Is_3 = \overline{K_3} \cdot \overline{K_2} \cdot K_1 \cdot \overline{K_0}$
- Detection for 5 (0101): $Is_4 = \overline{K_3} \cdot K_2 \cdot \overline{K_1} \cdot K_0$

K-maps and Derived Expressions

S_1S_0	Key
00	0
01	2
10	5
11	X

$S_1 = \overline{Is0} \cdot \overline{Is2}$

$S_0 = \overline{Is0} \cdot \overline{Is5}$

S_1S_0 \ Q_1Q_0	00	01	11	10
00	0	1	1	0
01	0	0	0	1
11	0	0	0	0
10	0	0	0	0

$D_1 = Q_1' = Q_0 \overline{S_1}\,\overline{S_0} + Q_1\overline{Q_0}\,\overline{S_1}S_0$

S_1S_0 \ Q_1Q_0	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	0	0	0	0
10	0	0	0	0

$D_0 = Q_0' = \overline{S_1}S_0$